

HATCom: A Self-Supervised Framework for Community Detection in Heterogeneous Attributed Temporal Networks

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Abstract. Community detection is a critical analytical task for uncovering structural patterns in complex, real-world networks. Incorporating heterogeneous nature, temporal dynamics and attributed networks from modern network data at once can provide useful and real-time information. This paper presents HATCom, a unified framework to detect communities in Heterogeneous, Temporal and Attributed Networks. The proposed model is built by integrating Relation-Time Co-Attention (RTCA) that captures dynamic interests of users and Graph Transformer Network (GTN) that captures soft meta-paths automatically. Furthermore, the system contains an Exponential Moving Average (EMA) based memory module to support temporal stability that captures the continuous evolution of node representations while supporting soft and overlapping community detection. To address scalability and supervision gaps, this system integrates Self-Supervised link prediction that trains the model without the need for manual labels. Experimental evaluation against DeepWalk, Node2Vec, and a simplified GTN variant demonstrates that HATCom achieves substantially stronger clustering quality, with a Silhouette Score of 0.6368 compared to 0.1130 for the strongest random walk baseline.

Keywords: Community Detection, Heterogeneous Graphs, Temporal Graph Learning, Self-Supervised Learning, Graph Attention, Graph Neural Networks

1 Introduction

Accurate community detection helps improve the user experience in complex applications such as recommendation systems, where predicting user groups improves engagement quality. Real-world data, like the MovieLens dataset, combines diverse node types, rich attributes, and temporal interactions (Harper and Konstan, 2015), yet traditional static graph methods fail to exploit this structure (Rani et al., 2023; Nair et al., 2022). We identify four concrete gaps that motivate HATCom:

1. **Attribute ignorance in static models:** While Node2Vec with Louvain are considered as a fundamental baseline for graph based Machine Learning models, it focuses on just graph structure and ignores temporal evolution and node attributes (Grover and Leskovec, 2016; Blondel et al., 2008; Jisha et al., 2021).
2. **Heterogeneity blindness in dynamic models:** CS-TGN captures temporal dynamics but processes all node types equally, not considering distinct semantic relations and hence degrading the embedding quality (Hashemi et al., 2023).
3. **Snapshot limitations:** DyHANE handles both heterogeneity and time but relies on coarse snapshot-based temporal discretisation (Martirano et al., 2024).
4. **Meta-path rigidity:** Most heterogeneous models require manually specified meta-paths, which is labour-intensive and domain-dependent (Behrouz and Hashemi, 2022).

HATCom addresses all four gaps through four design choices: a unified temporal heterogeneous graph combining users, movies, and genres; RTCA, which weights neighbours jointly by relation type and temporal recency; GTN, which discovers soft meta-paths automatically; and an EMA memory module that stabilises community assignments across time. Recent work confirms the importance of combining node attributes with relational structure (Deepthi and Swaminathan, 2025; Vennam et al., 2025), but neither method handles temporal dynamics, a gap HATCom fills. We evaluate against Deep-Walk + K-Means, Node2Vec + Louvain, and a Simplified GTN variant, and include an ablation study to quantify each component’s contribution.

2 Related Work

The trend in network representation learning has moved from static and homogeneous approaches to more dynamic and multi-relational approaches. This has been driven by the need to accommodate complex networks where there are different node types, edge semantics, and dynamic interactions.

2.1 Static Homogeneous Network Embedding

The combination of Node2Vec and Louvain are still widely used in community detection. Node2Vec uses biased random walks to learn node representation, which can be fed into Louvain’s modularity-based clustering (Grover and Leskovec, 2016; Blondel et al., 2008). Graph partitioning methods such as the enhanced Girvan-Newman algorithm (Jisha et al., 2021) also follow this divisive paradigm but similarly lack temporal and attribute awareness.

2.2 Dynamic and Temporal Graph Networks

To address the limitations of static snapshots, models like CS-TGN (Community Stability Temporal Graph Networks) introduce temporal stability constraints to temporal graph networks, ensuring smooth community evolution over time (Hashemi et al., 2023; Rani et al., 2023). However, they treat all node types uniformly, degrading the semantic quality of the embeddings. Furthermore, mod-

els like these suffer scalability issues due to continuous memory state updates across dense interactions.

2.3 Heterogeneous Network Embedding

As the complexity introduced by different entity types is observed, research such as DyHANE (Dynamic Heterogeneous Attributed Network Embedding) has been developed. This integrates structural and attribute information for dynamic community detection but suffers scalability and interpretability issues on dense multi-relational networks (Martirano et al., 2024). Similarly, models such as CS-MLGCN capture multi-level community structures via hierarchical convolution, yet require manually defined meta-paths and lack strong temporal modelling (Behrouz and Hashemi, 2022; Nair et al., 2022).

To address these gaps, HATCom integrates relation-aware attention, continuous timestamps, and automatic meta-path discovery. Graph attention network-based approaches (Pratiti and Srichandra, 2024) confirm that selectively weighting neighbour contributions significantly improves community detection quality.

3 Proposed Methodology

HATCom is a scalable, memory-efficient heterogeneous graph framework that performs community detection on a Heterogeneous Attributed Temporal Graph (HATG) built from the MovieLens 100K dataset (Harper and Konstan, 2015). The pipeline is shown in Fig. 1.

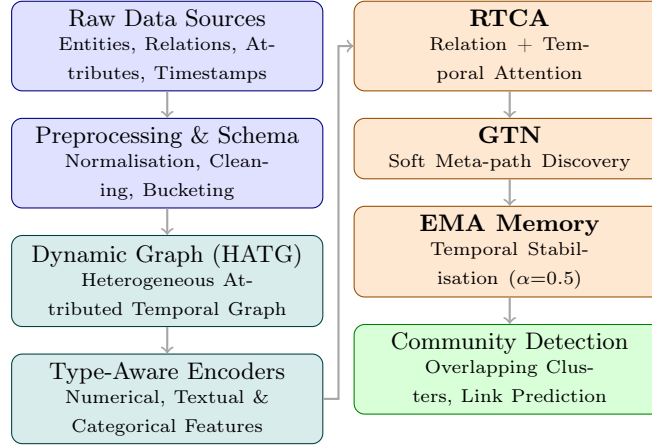


Fig. 1: HATCom pipeline.

3.1 Graph Construction and Data Preparation

The system uses the MovieLens dataset as a single Heterogeneous Attributed Temporal Graph (HATG) (Harper and Konstan, 2015), which natively supports multi-relational structures, temporal evolution, and rich node attributes. The graph consists of three node types: Users, Movies, and Genres, where Genre nodes enable message passing along *Movie* $\rightarrow$ *Genre* $\rightarrow$ *Movie* paths, allowing movies to share information through same genres. The full details of the graph construction, edge types, and feature encoding are provided in Section 4.

3.2 Type-Aware Attribute Encoding

Type-aware attribute encoders were implemented to handle various types of data while also handling attribute diversity. These different encoders combine different types of attributed data, such as numerical and textual attributes, into a unified embedding space. This enables the model to process users, movies and genres in a unified and consistent manner without losing their original semantic meaning.

3.3 Relation-Time Co-Attention (RTCA)

Instead of processing the temporal data linearly, the RTCA module aggregates the neighbour information by evaluating these two dimensions simultaneously:

1. **Relation Semantics:** The specific type of relation between two nodes (Martirano et al., 2024).
2. **Temporal Recency:** The module integrates temporal awareness by representing timestamps as edge attributes. It uses an attention mechanism to assess the contribution of interactions, resulting in an $O(1)$ temporal encoding efficiency with time difference calculations.

The attention score for an edge (i, j) with relation type r and timestamp t is computed as:

$$e_{ij} = \text{LeakyReLU}(\mathbf{W}_a \cdot [\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_r \parallel \phi(t)]) \quad (1)$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(j)} \exp(e_{kj})} \quad (2)$$

$$\mathbf{h}_j^{\text{RTCA}} = \sum_{i \in \mathcal{N}(j)} \alpha_{ij} \cdot \mathbf{h}_i \quad (3)$$

where $\mathbf{e}_r$ is the relation type embedding and $\phi(t)$ is the temporal encoding defined as:

$$\phi(t) = \exp\left(-\frac{t_{\max} - t}{\tau}\right) \quad (4)$$

where $t_{\max}$ is the maximum timestamp and τ is the decay constant (set to 30 days).

3.4 Graph Transformer Network (GTN)

To overcome the complexity of manual meta-paths, HATCom uses a Graph Transformer Network. This layer is responsible for:

- Capturing underlying dependencies automatically, which may not be captured by manual rules.
- Improving the interpretability of the learned embeddings by using edge-wise attention (Martirano et al., 2024).

The multi-channel aggregation is formulated as:

$$\mathbf{H}^{\text{GTN}} = \left\|_{k=1}^K \sum_{j \in \mathcal{N}(i)} \alpha_{ij}^{(k)} \cdot \mathbf{W}^{(k)} \mathbf{h}_j \right. \quad (5)$$

where $K = 2$ is the number of channels, $\alpha_{ij}^{(k)}$ are channel-specific attention weights, and $\parallel$ denotes concatenation.

3.5 Memory Module and Temporal Stability

A memory module is implemented to ensure temporally stable communities. It works by capturing the evolution of node embeddings over time, preventing any sudden changes in structure that were temporary. Hence, it helps in discovering stable overlapping communities (Hashemi et al., 2023). The EMA-based memory update is defined as:

$$\mathbf{M}_t = \alpha \cdot \mathbf{H}_t + (1 - \alpha) \cdot \mathbf{M}_{t-1} \quad (6)$$

where $\alpha = 0.5$ is the memory update rate, $\mathbf{H}_t$ is the current GTN output, and $\mathbf{M}_{t-1}$ is the previous memory state.

3.6 Self-Supervised Learning

The model is trained using self-supervised link prediction on user–movie edges. The model learns high-quality embeddings without the need for ground-truth labels. The training objective is:

$$\mathcal{L} = - \sum_{(u,v) \in \mathcal{E}^+} \log \sigma(\mathbf{h}_u \cdot \mathbf{h}_v) - \sum_{(u,v) \in \mathcal{E}^-} \log(1 - \sigma(\mathbf{h}_u \cdot \mathbf{h}_v)) \quad (7)$$

where $\mathcal{E}^+$ and $\mathcal{E}^-$ are positive and negative edge sets respectively (Nair et al., 2023). Training uses 20,000 positive user–movie edges subsampled from the full ratings set, with an equal number of negative edges sampled from non-existing user–movie pairs. The dataset is split 80/20 for training and validation.

4 Experimental Setup

4.1 Dataset and Graph Construction

We use the MovieLens 100K dataset (Harper and Konstan, 2015), which contains 943 users with demographic attributes, 1,682 movies with titles, release dates, and genre labels, 100,000 timestamped ratings, and 18 genre categories providing the heterogeneous nodes, temporal data, and attribute diversity needed for HATG construction. The heterogeneous graph $G = (V, E)$ is constructed with $V = V_u \cup V_m \cup V_g$ (2,643 nodes total) and four edge types totalling 1,847,945 edges: *rates* (user-to-movie, with rating and timestamp), *has_genre* (movie-to-genre), *user_similarity* (user-to-user via Pearson correlation), and *shares_genre* (movie-to-movie via cosine similarity > 0.5). Genre nodes enable message passing along *Movie*→*Genre*→*Movie* paths, capturing co-genre relationships structurally. The graph is visualised in Fig. 2.

4.2 Feature Encoding

The encoding of user features (age, gender, occupation) into a 256-d vector is done with MLP based on sentence-transformer embeddings (all-MiniLM-L6-v2); the encoding of movie titles and genre vectors into 448-d vectors and the encoding of genre nodes into 384-d vectors. Graph processing is done with all features projected to a common 128-d space using MLP Encoder modules.

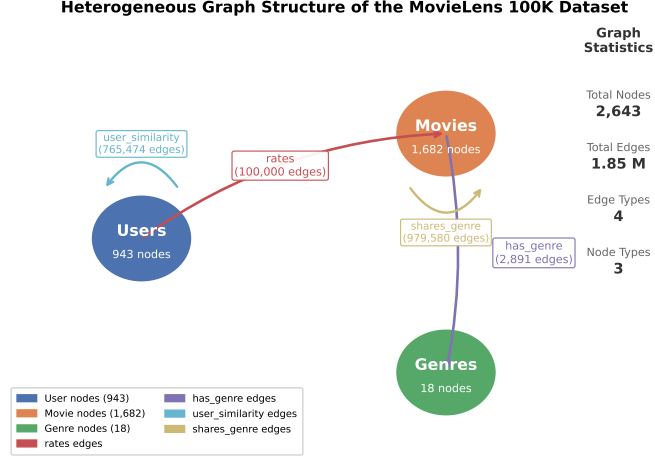


Fig. 2: Heterogeneous graph: 3 node types, 4 edge types, 1.85M edges, 2,643 nodes.

4.3 Evaluation Metrics

We present Silhouette Score, Calinski-Harabasz (CH) Index, Davies-Bouldin (DB) Index, and Modularity (Q) for community detection quality, Normalised Mutual Information (NMI) for temporal stability between snapshots, and AUC-ROC for link prediction.

4.4 Implementation Details

PyTorch 2.1.0 is used for the implementation of HATCom. Key hyperparameters: RTCA attention dimension 128; GTN channels $K=2$; EMA rate $\alpha=0.5$; Adam optimiser, learning rate 10^{-3} ; batch size 512 edges; 5 training epochs; temporal decay $\tau=86400 \times 30$ seconds; 5 uniform temporal snapshots. Training is performed on a subsample of 300,000 edges for message passing (all nodes retained); of these, 20,000 positive user-movie edges are drawn as the self-supervised training objective, with an equal number of negative pairs sampled from non-existing user-movie pairs. The remaining edges serve as the message-passing context. Experiments are run on a single 16 GB NVIDIA GPU.

5 Results and Analysis

5.1 Community Detection Performance

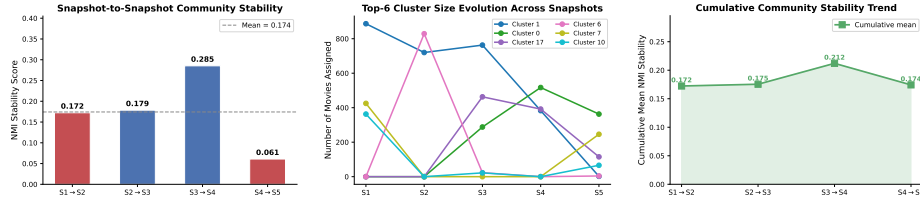
On learned embeddings, HATCom identifies 18 communities across 1,682 movies via K-Means. $K=18$ was chosen to match the number of genre categories in MovieLens, enabling direct evaluability of cluster-genre correspondence without supervising the clustering process itself. Table 1 summarises the clustering metrics.

Well-separated communities are validated by a Silhouette Score of 0.6368 (greater than the 0.5 threshold). The CH Index value of 41,157 indicates compact and discrete clusters, the DB Index value of 0.5284 indicates a low degree of inter-cluster overlap, and the Modularity value of 0.8579 confirms a high degree of intra-community connectivity. A cluster purity of 0.4479 is attained without

Table 1: Community Detection Performance of HATCom

Metric	Value	Metric	Value
Silhouette Score	0.6368	Calinski-Harabasz Index	41157.85
Davies-Bouldin Index	0.5284	Modularity (Q)	0.8579
Number of Clusters	18	Average Conductance	0.0604
Avg. Cluster Purity	0.4479	Avg. Temporal Stability	0.3725

genre supervision, indicating that the embeddings capture semantically meaningful structure; the cluster with the highest genre purity (Cluster 16, Comedy-dominant, purity 0.64) and the most mixed (Cluster 2: Action 38%, Thriller 30%, Comedy 28%) both represent genuine movie similarity rather than label leakage. Cluster sizes range from 13 to 366 movies, consistent with the uneven distribution of genres in MovieLens.



(a) Snapshot NMI stability. Peak at S3→S4 (0.285); drop at S4→S5 due to skewed edge distribution. (b) Top-6 cluster sizes across snapshots. EMA stabilises from S3 onward. (c) Cumulative mean NMI. Peak at S3→S4 (0.212); final value 0.174.

Fig. 3: Temporal stability analysis across five snapshots.

Temporal stability is evaluated through NMI between consecutive snapshot assignments derived over five snapshots, as shown in Fig. 3. Stability increases between S1→S2 (0.172) to a peak at S3→S4 (0.285) as the EMA memory accumulates historical context, then decreases drastically between S4→S5 (0.061) because the final snapshot contains 286,791 edges versus fewer than 5,000 in each earlier snapshot, a structural shift that is beyond the capacity of the EMA to adjust. The top-6 cluster sizes show corresponding volatility at S2 (sparse interactions) and stability from S3 onwards. Finally, the cumulative mean NMI peaks at S3→S4 (0.212) and levels at 0.174 after the anomalous final snapshot.

5.2 Genre Classification and Link Prediction

Table 2: Downstream task performance

(a) Genre Classification (KNN, $k=5$)

Metric	Value
Classification Accuracy	68.15%
Training Samples	1,343
Test Samples	336
Genre Classes	18

(b) Link Prediction Performance

Metric	Value
Final Val AUC-ROC	0.7950
Final Val Accuracy	71.82%
Final Train Loss	0.5934
Final Val Loss	0.5821

Tables 2a and 2b summarise the downstream task performance.

Genre classification accuracy of 68.15% against a 5.9% random baseline confirms that the embeddings encode rich semantic content. The RTCA mechanism effectively aggregates heterogeneous edge types, and the GTN layers identify meaningful meta-paths between similar films. Link prediction converges stably within 5 epochs (train loss 0.5934, val loss 0.5821) with no overfitting, and the validation AUC of 0.7950 confirms that the self-supervised objective produces high-quality pairwise representations.

5.3 Baseline Comparison

We compare against three baselines. **DeepWalk + K-Means** treats the graph as homogeneous, using uniform random walks ($p=q=1$) with K-Means ($K=18$). **Node2Vec + Louvain** uses biased random walks with modularity-based clustering (Grover and Leskovec, 2016; Blondel et al., 2008). **Simplified GTN + K-Means** runs only the GTN on projected node features, omitting RTCA and the memory module, isolating the contribution of relation-time attention and temporal stabilisation. DeepWalk and Node2Vec are evaluated on a sparse KNN graph ($K=10$, cosine similarity) due to memory constraints during random walk pre computation; HATCom and Simplified GTN use the full heterogeneous HATG with 300,000-edge subsampling. NMI and ARI are excluded from the primary comparison because random walk baselines traverse explicit *shares_genre* edges, which directly encode genre co-membership and inflate label-agreement metrics without reflecting genuine representation quality.

Table 3: Comparison of HATCom against baseline methods on MovieLens 100K.

Method	Silhouette $\uparrow$	CH Index $\uparrow$	DB Index $\downarrow$	Modularity $\uparrow$
DeepWalk + K-Means	0.1130	63.94	2.2492	0.6790
Node2Vec + Louvain	0.0876	58.65	2.6539	0.7238
Simplified GTN + K-Means	0.2994	4571.99	1.0673	0.7818
HATCom (Proposed)	0.6368	41157.85	0.5284	0.8579

Table 3 shows a clear performance gap. The most informative comparison is against Simplified GTN, which shares the same graph construction and feature encoding pipeline: Silhouette Score improves from 0.2994 to 0.6368 and CH Index from 4,571 to 41,157 when RTCA and the memory module are added. This gap cannot be attributed to architectural complexity alone; the Simplified GTN already accesses the full heterogeneous graph via GTN layers. The improvement reflects RTCA’s ability to distinguish a recent high-confidence rating from an old weak one, which is critical in MovieLens where user preferences evolve over years and edge semantics vary significantly. Random walk baselines score poorly on geometric metrics despite their access to *shares_genre* edges, confirming that embedding quality rather than the clustering algorithm is the bottleneck, the Node2Vec + Louvain DB Index of 2.65 despite using a modularity-maximising clusterer makes this point directly. All methods achieve relatively high modularity (0.67–0.86), reflecting inherent community structure in the graph; HAT-

Com’s advantage is that it simultaneously achieves high modularity *and* strong geometric separation, a combination the baselines do not achieve.

5.4 Visualisation and Analysis

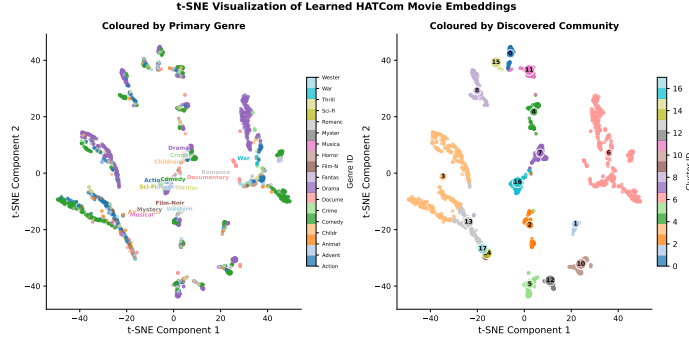


Fig. 4: t-SNE of movie embeddings by genre (left) and K-Means cluster (right).

Fig. 4 shows t-SNE projections of all labelled movie embeddings, coloured by primary genre (left) and K-Means community assignment (right). The strong structural correspondence between the two panels confirms that the self-supervised pipeline produces semantically coherent representations without label supervision. Well-separated clusters emerge for Action/Adventure, Drama/Romance, and Comedy; movies near cluster boundaries reflect genuine multi-genre content.

Fig. 5 shows RTCA attention weights by edge type and temporal bin (rows normalised independently). T5 (most recent bin) dominates for all edge types, confirming that the exponential decay encoding successfully focuses the model on recent interactions; *rates* edges show the steepest recency gradient, consistent with their role as the primary supervision signal. Across edge types, the model learns a useful priority ordering without any manual specification: *rates* edges contribute most to final embeddings, followed by *user_similarity* and *has_genre*, with *shares_genre* contributing least, a ranking that matches the relative informativeness of each relation type for community assignment. T5 captures 73.3% of total attention weight across all edge types, quantifying the strength of the recency effect.

5.5 Ablation Study

We execute five configurations: Full HATCom, no RTCA, no GTN, no Memory, and Base Graph (projected features only), fixing all hyperparameters and edge subsampling. Table 4 reports results across five configurations. Note that all ablation runs use a single-snapshot sub graph for controlled comparison; values therefore differ from Table 1, which reflects full temporal integration with EMA.

The three modules make their contribution differently. The maximum decrease in clustering is obtained when the memory module is removed (Silhouette 0.2456), confirming the role the memory module plays in averaging temporal fluctuation across snapshots. The following most damaging deletion is of RTCA

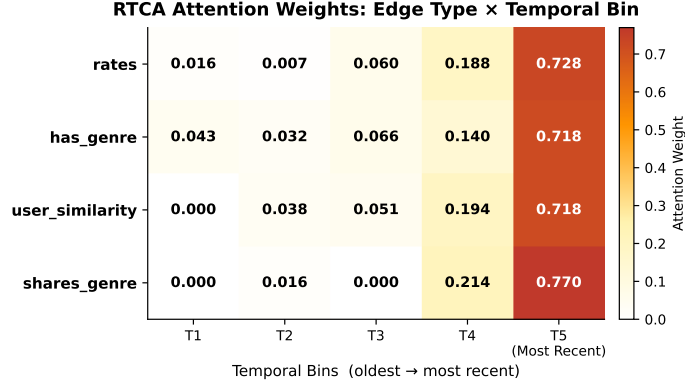


Fig. 5: RTCA attention weights by edge type and temporal bin (rows normalised). T5 dominates, confirming recency bias.

Table 4: Ablation Study Results

Configuration	Silhouette	CH Index	DB Index	Link AUC	Genre Acc
Full Model	0.6368	24354.05	0.5548	0.7950	0.6815
w/o RTCA	0.3270	2677.96	1.5233	0.8013	0.5357
w/o GTN	0.3993	3959.04	0.9822	0.8000	0.7054
w/o Memory	0.2456	2167.93	1.5772	0.7932	0.5030
Base Graph	0.1694	988.76	1.5273	0.5923	0.5893

(Silhouette 0.3270), reflecting the cost of treating a recent high-confidence rating identically to a decade-old weak one. The w/o GTN result is marginally smaller: there is a decrease in clustering quality and a marginally higher accuracy of genre classification (0.7054 vs. 0.6815), showing that soft meta-path aggregation of GTN does indeed increase global community coherence at a tolerable cost to genre discrimination at the movie level, partially compensated by benefits in clustering. Link prediction AUC does not vary significantly across partial configurations (0.79 to 0.80), confirming that local pairwise similarity is captured by graph learning alone; the base graph drop to 0.5923, showing that message passing is required to obtain community-quality representations.

6 Conclusion and Future Work

HATCom addresses the gap of community detection through bringing heterogeneity, temporal dynamics and node attributes together into one self-supervised model. The model with RTCA, GTN, and EMA-based memory module has a Silhouette Score of 0.6368 and Calinski-Harabasz Index of 41,157.85 on MovieLens 100K, which is very much higher than all the baselines. The ablation study validates the fact that all the components add value, but the most important component to the quality of clustering is the memory module and RTCA.

The existing shortcomings are the evaluation done only on a single dataset, the need to retrain the system on new interactions, and the lack of interpretability of the learned meta-paths. The next steps in the work include incremental training in streaming environments, generalising to citation and biological net-

works, and replacing link prediction pretraining with graph contrastive learning to produce more informative representations.

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